EXPERIMENT-3 (STUDY)

STUDY OF NETWORK CABLES AND CRIMPING WITH RJ45 CONNECTOR

Aim:

To study different types of network cables & to prepare a straight-through & to prepare cross-over Ethernet cable by crimping RJ45 connectors.

THEORY:

1. Twisted pair cable:

UTP (unshielded) & STP (shielded) used in LANs.

Categories:

→ cat5
→ cable
→ cat 6
→ cat 6a.

2. Coaxial cable:

Used in cable TV & other ethernet (10Base 10Base 5).

3. Fiber optic cable:

Transmits data as light pulses. Single mode (long distance) & multi mode (short distance)

Very high bandwidth.

→ RJ45 is the standard connector for ethernet cables.

Two wiring standards exists.

- TIA/EIA 568A:

Green-white, green, orange-white, Blue, Blue-white, Orange, Brown-white, Brown.

- TIA/EIA 568B:

Orange-white, orange, Green-white, Blue, Blue-white, Green, Brown-white, Brown.

RESULT:

Both straight through & cross-over ethernet cables were successfully crimped & verified using a cable tester. All 8 pins showed proper continuity.